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Group 8

FINANCIAL ENGINEERING
Assignment n°4 - FE

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0 Introduction

This report presents the analysis and solutions for Assignment 4 on Structured Products. The project explores advanced pricing methodologies, starting with the calculation of the upfront payment for a Certificate linked to an equity basket , followed by the pricing of digital options accounting for the volatility smile. Next, we implement and compare numerical pricing techniques—including Fast Fourier Transform (FFT), quadrature, and Monte Carlo simulations—for normal mean-variance mixture models, such as the Normal Inverse Gaussian (NIG) model. Finally, the report concludes with a practical case study on volatility surface calibration using S&P 500 data.

1. Certificate Pricing

On February 15, 2008, Bank XX issues a certificate structured as a swap between two counterparties. Party A pays a floating leg (Euribor 3m + 130bps, quarterly, Act/360) plus a protection payment $(1-P)\cdot B_T$ at maturity, where $P = 95\%$ is the recovery rate and B_T is the discount factor at maturity $T = 5$ years. Party B pays an upfront $X\%$ of the notional and, at maturity, a coupon equal to $\alpha \cdot (S(T) - P)^+$, where $\alpha = 110\%$ is the participation coefficient and $S(T)$ is the equally-weighted basket performance of ENI and AXA.

The pricing equation follows from the no-arbitrage condition:

$$\mathrm{PV}_{\mathrm{Float}} + \mathrm{PV}_{\mathrm{Spread}} + \mathrm{PV}_{\mathrm{Protection}} = X + \mathrm{PV}_{\mathrm{Coupon}}$$

where $\mathrm{PV}_{\mathrm{Float}} = 1 - B_T$, $\mathrm{PV}_{\mathrm{Spread}} = s_{\mathrm{spol}} \sum_i \delta_i B_i$ (with $s_{\mathrm{spol}} = 1.3\%$), and $\mathrm{PV}_{\mathrm{Protection}} = (1-P)\cdot B_T$. The coupon present value is estimated via Monte Carlo simulation ($N = 10^6$ paths) of the basket under the risk-neutral measure, using the correlated GBM dynamics of the two underlyings (ENI: $S_0 = 12.3$, $\sigma_1 = 20.1\%$, $d_E = 3.2\%$; AXA: $S_0 = 22.1$, $\sigma_2 = 18.3\%$, $d_{CS} = 2.9\%$; $\rho = 49\%$), discounted at the bootstrapped zero curve. The resulting theoretical upfront is:

$$\boxed{X = 9.940\% \text{ of the notional}}$$

2. Digital Option Pricing

A cash-or-nothing digital call on the Eurostoxx 50 is priced as of February 15, 2008, with the following contract specifications: notional $N = 10$ Mln EUR, digital payoff = 5% of notional (Cash = 500,000 EUR), ATM forward strike $K = F = 3764.39$, and expiry $T = 1$y (Act/365). Three pricing approaches are compared.

Black model. Under flat ATM volatility ($\sigma_{\mathrm{ATM}} = 23.72\%$), the digital is priced as $\mathrm{Price}_{\mathrm{Black}} = \mathrm{Cash} \cdot B(0,T) \cdot N(d_2)$, yielding EUR 217,650.79. Since the strike equals the forward, $d_1 = -d_2 = 0.1186$ and $N(d_2) = 0.4528$.

Smile correction (analytic). The Eurostoxx 50 implied volatility surface exhibits a pronounced negative skew ($\partial\sigma/\partial K = -0.000061$), meaning lower strikes carry higher implied volatility. Ignoring this leads to mispricing. The smile-corrected price is obtained via a first-order adjustment:

$$\mathrm{Price}_{\mathrm{Smile}} = \mathrm{Price}_{\mathrm{Black}} - \mathrm{Cash} \cdot \mathcal{V} \cdot \frac{\partial\sigma}{\partial K}$$

where $\mathcal{V} = B(0,T)\, n(d_1)\, \sqrt{T}\, F = 1433.61$ is the vega of the vanilla call. The correction amounts to −EUR 43,445.91, giving EUR 261,096.71.

Call spread (numerical verification). The digital is replicated as $\frac{C(K-\varepsilon)-C(K+\varepsilon)}{2\varepsilon}$ using Black prices at $K \pm 1$, each evaluated at the corresponding implied volatility. The result, EUR 261,096.69, confirms the analytic correction to high precision.

The Black model underestimates the digital price by 8.69% of the payoff. This bias arises because the negative skew raises the risk-neutral probability of finishing in the money for a downward-sloping smile, a correction invisible to the flat-vol Black formula.

1 Facultative: Pricing with a given characteristic function

Starting from the following characteristic function:

$$\phi^c(u) = \frac{1}{(1 - i\frac{u}{p^+})(1 + i\frac{u}{p^-})} e^{i\mu u} \tag{1}$$

this point of the Assignment asks us to compute the call price C with the Lewis formula using different approaches. We also write directly the Lewis formula in order to have a clear connection with what we will comment in the following points of the report:

$$C(x) = F_0 B(t_0, t) * (1 - \frac{e^{-\frac{x}{2}}}{2\pi} \int_{-\infty}^{+\infty} d\xi \phi(-\xi - \frac{i}{2}) \frac{e^{-i(\xi + \frac{i}{2})x}}{\xi^2 + \frac{1}{4}}) \tag{2}$$

Given the characteristic function, in order to find μ, we applied, as requested, the following martingale condition on the characteristic function:

$$1 = \phi_f(-i) \tag{3}$$

and after the calculation we obtained the following formula for μ:

$$\mu = ln((1 - \frac{1}{p^+})(1 + \frac{1}{p^-})) \tag{4}$$

In the following part of the report we discuss the results obtained varying the approaches:
a. Compute C with quadrature:
In this point of the Assignment we computed the price of a Call option handling the integral of Lewis Formula using a quadrature method. The main thing to underlie, is the fact that we used an upper limit for integration equal to 20. At first we decided to use 100 to be sure to have a really close result to the one given by the other methods, but then, given the fact that the exponential part of the integrand decays really fast, we decided to rerun the code in order to find the lowest upper limit that could give us a closer results to the ones obtained with 100. We also did the same passages for the discretization steps that in the code are reported with the variable N_z (in particular we arrived to 100). We did this passages for both terms since if we just reduced the number of steps N_z we would have obtained a closer result only using a larger number of steps N_z. This is something really important to do in order to obtain a really reliable result without spending a lot over the necessary on the computation side. Here the prices obtained for the given x in the Assignment:

x	C
-5.223%	$1,563.66$€
0%	$1,605.05$€
15%	$1,730.06$€

Table 1: Prices with quadrature

The computed Call prices exhibit a strictly increasing trend with respect to the log-moneyness x. Assuming the standard convention $x = \ln(F_0/K)$, a negative moneyness ($x = -5.223\%$) implies the strike price K is higher than the forward price F_0. Therefore, the option is Out-Of-The-Money (OTM), resulting in the lowest premium. Conversely, $x = 0$ represents the At-The-Money (ATM) forward scenario, while a strongly positive moneyness ($x = +15\%$) indicates the option is deeply In-The-Money (ITM), which logically justifies the highest option price. This behavior from our point of view aligns with theoretical financial expectations.
b. Compute C with residual's technique:
The residual's technique is the analytical approach in order to calculate the price of a Call, for this reason the results given by this approach will be our benchmark. In the following table the results given by this approach:

x	C
-5.223%	$1,563.67$€
0%	$1,605.05$€
15%	$1,730.06$€

Table 2: Prices with residuals

c. Compute C via Monte Carlo:
This point of the Assignment asks us to compute the prices C using Monte Carlo simulations, leaving a hint to compute the Cumulative Distribution Function related to the Characteristic Function given. Starting from this hint, we noticed that we can see the characteristic function as the product of three parts. The first part is $\phi^c_\mu(u) = e^{i\mu u}$, which can be seen as the characteristic function of the constant μ. The other two parts: $\phi^c_{E_1}(u) = \frac{1}{1-i\frac{u}{p^+}}$ and $\phi^c_{E_2}(u) = \frac{1}{1+i\frac{u}{p^-}}$ are really both similar to the characteristic function of an Exponential random variable $Y \sim \text{Exp}(\lambda)$, ie $\phi_Y(u) = \frac{\lambda}{\lambda - iu}$. So, remembering the fact that the product of three characteristic functions is the characteristic function of a random variable defined as the sum of the three independent random variables related to the three characteristic functions respectively, we can define a random variable Y as follows: $Y = \mu + E_1 - E_2$, where $E_1 \sim \text{Exp}(p^+)$ and $E_2 \sim \text{Exp}(p^-)$ and μ is a constant as written above. This makes the structure of the Monte Carlo simulation, because we know that *rand* in MATLAB gives back an array of uniformly distributed random variables and knowing the fact that if $X \sim \text{Unif}(0,1)$, then $-ln(X) \sim \text{Exp}(1)$, we can directly treat E_1 and E_2 simulating the paths for both indipendently (as the characteristic function relation of Y suggests). The results are the following (1Mln simulations):

x	C
-5.223%	$1,521.25$€
0%	$1,562.68$€
15%	$1,687.94$€

Table 3: Prices with MC simulations

The results obtained are different from the one given by the other techniques. From our point of view, this may be related to the statistical noise that the Monte Carlo method has in its nature. Apart from this particular thing, the path of the prices is exactly the one seen in the quadrature approach which, from my point of view, is the key thing that had to be present also in this approach. **d. Compute C via FFT:**
In order to compute the prices C in the FFT approach we had to define a value for the steps dz. In order to be sure to obtain the best results possible we started with a value $dz = 0.001$. Then, once we obtained prices really close to the quadrature and residual approaches ones, we started to improve the parameters from a code point of view, lowering the computation work. So, by trying we lowered dz to 0.1, obtaining approximately the same prices obtained with $dz = 0.001$. We had to adjust these components in order to obtain a good result without spending so much on the computation side. Of course, the same argument has to be applied to choosing M, in class it was recommended to choose a number around 10 and 15, also by modifying these values, we tried to choose the lowest value that could give us the best result possible (trade off: results-computational working). Seeing that for $M = 10$ the results are really close to the other approaches, we chose this value. The other parameters are chosen following the hypothesis given in class by the Professor. Here the results:

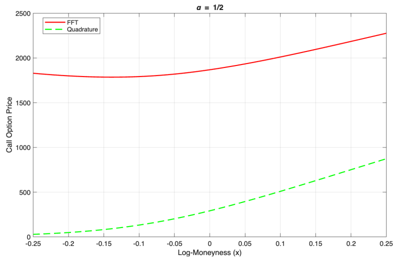
x	C
-5.223%	$1,563.68$€
0%	$1,605.05$€
15%	$1,730.06$€

Table 4: Prices with FFT approach

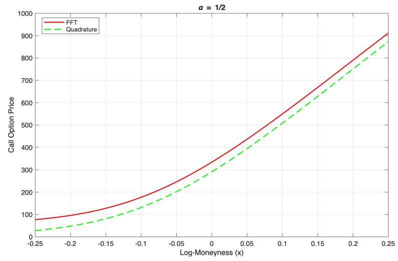
The results are practically the same to the ones of the quadrature, the same comments apply.

2 Pricing a Call option using FFT, Quadrature, and Monte Carlo

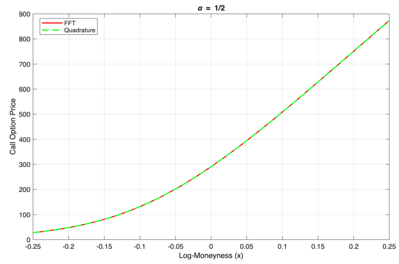
In point 4 of the Assignment is asked to compute the prices C according to a Normal Mean Variance Mixture (NMVM) with $\alpha = \frac{1}{2}$ (NIG model).
FFT approach:
For the FFT approach, we had to decide an adequate value for M and dz. The main path to select these parameters is the one presented in the part on the FFT approach in the exercise 3 of this Assignment. In order to do the analysis requested by the Assignment we take the results obtained with the quadrature approach and we analyze the error with respect to that results:



(a) Case M=8



(b) Case M=10



(c) Case M=12

Figure 1: Comparing FTT Vs Quadrature approaches varying M

The analysis show that using a value of M higher or equal than 12, in the case of $dx = 0.01$ gives us a Price path varying the strikes really close to the one given by the Quadrature approach which we chose as a benchmark. This analysis led us to choose $M = 12$ in order to obtain a good result without having to suffer from a computational point of view. Another key thing to underlie in this part of the assignment is the discussion on the choosing the right parameters: some relations are right from the theory, like the following: $N = 2^M$, $dzdx = \frac{2\pi}{N}$. Given the fact that dx is given by the Assignment and M is chosen by the analysis done in the previous lines, the only parameter that we had to handle is z. We chose to handle it using the formula present also in the Excel file for the first value of z:

$$z_1 = -\frac{(N-1)dz}{2} \tag{5}$$

and then we defined the other points of z proceeding summing dz.

Quadrature approach

We handled the quadrature approach exactly as we handled it in the part 3 of the Assignment, so we started with a very large value for the upper limit of integration, we started from 1000 and in the final we took an upper limit equal to 20. This was possible thanks to the fact that the exponential of the integrand decays really fast. As explained also in point 3, this enables us to lower the length of the discretization steps of the integral: we started from a number of steps equal to 10,000 and in the end we were able to reduce it to 20 preserving the values of the results obtained with the parameters' values used at the start. This practice has to be done in order to reduce the computational work.

Monte Carlo approach:

For the Monte Carlo approach we had to handle in the right way the positive random variable G and to do so, we followed step by step the formulas seen in class. Once we had the simulated log-returns we found the prices at maturity of the Forward, computed the payoffs, their mean and discounted them to get the prices. In order to be sure of the computations done with the parameters that we are handling we have done check as requested by the Assignment on the moments. This test showed not big discrepancies between the analytical and empirical moments.

Final results and comments: In the following plots we show the comparisons between the approaches presented in the previous points and also a comparison between the NIG method and the one with $\alpha = \frac{1}{3}$ using the values of the parameters discussed previously:

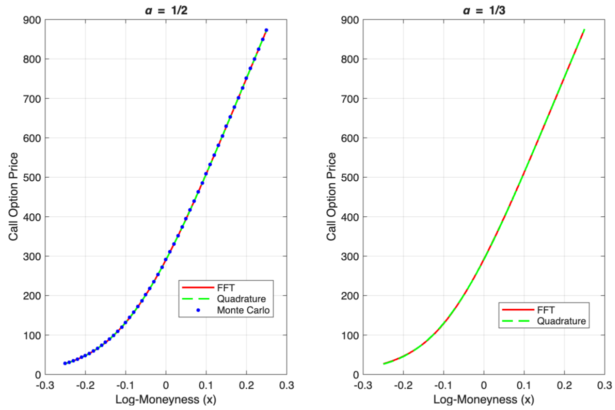


Figure 2: Verification of approaches

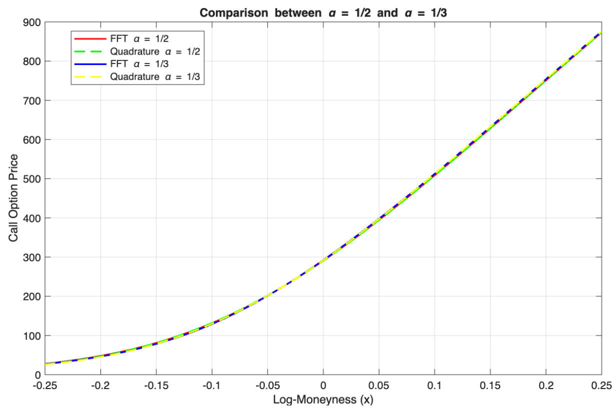


Figure 3: Varying α

The results presented from the plots are optimal from our point of view. The comparison between the approaches show that varying the approach used (FFT, MC, quadrature), the prices computed do not change really much and most importantly, they follow the same path varying the strike. Also the comparison between $\alpha = \frac{1}{2}$ and $\alpha = \frac{1}{3}$ shows that the two methods present practically the same path in the pricing varying the value of the strike. From our point of view this is a good news, which shows also the fact that varying the methods ($\alpha = \frac{1}{2}$ or $\alpha = \frac{1}{3}$) the parameters that we chose manually, eg z_1 and M, give an optimal result. This was not granted, since from a statistical point of view we expected to have to increase the chosen parameters. As observed in the plots, the call option prices increases as the log-moneyness x grows. This is expected by the definition of log-moneyness: $x = ln(\frac{F_0}{K})$, so an increase in the value of log-moneyness corresponds to a lower strike K which leads to increasing the value of the call option.

3 Volatility surface calibration

This section presents the global calibration of a Normal Mean-Variance Mixture (NMVM) model with $\alpha = \frac{2}{3}$ on the S&P 500 implied volatility surface. The forward price F_0, the discount factor $B(t_0, T)$, and the time to maturity T are taken directly from the values computed in Section 2.

Model

The log-return of the underlying follows a NMVM process. The Laplace exponent of the mixing subordinator is:

$$\ln \mathcal{L}(\omega) = \frac{T}{\kappa} \cdot \frac{1-\alpha}{\alpha} \left[1 - \left(1 + \frac{\kappa\sigma^2 \, \omega}{1-\alpha}\right)^{\alpha}\right] \tag{6}$$

where $\kappa > 0$ controls the kurtosis of the distribution, $\sigma > 0$ is the volatility parameter, and η governs its skewness. The drift μ is determined by the martingale condition $\varphi(-i) = 1$, which yields:

$$\mu = -\ln \mathcal{L}(\eta) \tag{7}$$

The characteristic function of the log-return, with the martingale condition already applied, is:

$$\varphi(\xi) = e^{i\xi\mu} \cdot \exp\left[\ln \mathcal{L}\left(\frac{1}{2}\left(\xi^2 + i(1+2\eta)\xi\right)\right)\right] \tag{8}$$

The call price is then computed via the Lewis formula:

$$C(x) = F_0 \, B(t_0, T) \left(1 - \frac{e^{-x/2}}{2\pi} \int_{-\infty}^{+\infty} \frac{\varphi\left(-(z + \frac{i}{2})\right)}{z^2 + \frac{1}{4}} \, e^{-ixz} \, dz\right) \tag{9}$$

where $x = \ln(F_0/K)$ is the log-moneyness. The integral in (9) is evaluated numerically via FFT using the same grid parameters adopted in Section 4 ($M = 14$, $dx = 0.01$).

Calibration procedure

The three model parameters $\boldsymbol{\theta} = (\kappa,\, \eta,\, \sigma)$ are estimated by minimising the sum of squared errors (SSE) between the model call prices and the market call prices obtained by feeding the observed implied volatility surface into the Black formula:

$$\hat{\boldsymbol{\theta}} = \arg\min_{\boldsymbol{\theta}} \sum_{i=1}^{N_K} \left[C^{\text{model}}(K_i;\, \boldsymbol{\theta}) - C^{\text{mkt}}(K_i)\right]^2 \tag{10}$$

The optimisation is carried out with MATLAB's `fmincon` solver, subject to the following box constraints:

$$\kappa \in [0.1,\ 1], \qquad \eta \in [-50,\ 50], \qquad \sigma \in [0.1,\ 1] \tag{11}$$

and the nonlinear integrability constraint, which ensures that the Laplace exponent in (6) is well defined at $\omega = \eta$:

$$\eta + \frac{1-\alpha}{\kappa\, \sigma^2} \geq 0 \tag{12}$$

Constraint (12) guarantees that the argument of the power function in (6) is strictly positive, avoiding branch-cut issues in the complex plane. The lower bound $\sigma > 0$ is imposed because σ enters the model only through σ^2, making its sign non-identifiable; restricting it to be positive removes this redundancy.

$\hat{\kappa}$	$\hat{\eta}$	$\hat{\sigma}$
0.2548	25.2702	0.1374

Table 5: Calibrated NMVM parameters ($\alpha = 2/3$).

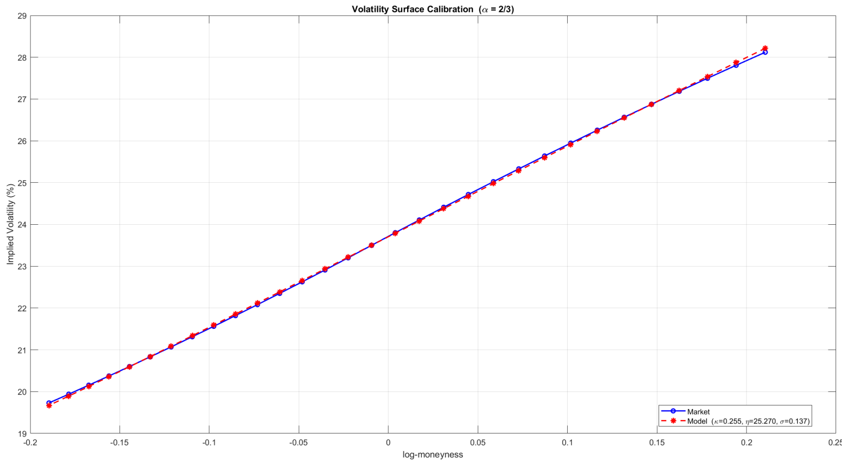


Figure 4: Market vs. model implied volatility surface after calibration ($\alpha = 2/3$, $\hat{\kappa}$ =[0.2548], $\hat{\eta}$ =[25.2702], $\hat{\sigma}$ =[0.1374]).

Results

The calibrated parameters are reported in Table 5.

Figure 4 compares the market implied volatility smile with the one produced by the calibrated model, both plotted against the log-moneyness.

From Figure 4 we can observe that the calibrated NMVM model is able to reproduce the main features of the S&P 500 volatility smile. Plotting the implied volatility against log-moneyness x is particularly natural in this context, since the Lewis formula (9) is expressed directly in terms of x, and the FFT grid is built uniformly in this variable. The smile exhibits the well-known negative skew of equity index options: implied volatility increases as x grows (i.e. as the strike K decreases relative to F_0), reflecting the market's higher pricing of downside risk. The parameter η controls this asymmetry through the term $(1+2\eta)$ in the argument of the characteristic function (8), while κ governs the overall curvature of the smile and σ sets its level. The quality of the fit confirms that the NMVM framework with $\alpha = \frac{2}{3}$ provides sufficient flexibility to match the observed market surface.